



The Building Knows First

Forecasting Campus-Space Abandonment from Desk-Booking and Swipe-Card Signals

Petra Umbile · Anneke Tolan — School of Emergent Priorities

Background

Facilities teams learn a building has gone quiet only once a quarterly utilisation report says so. Do desk-booking and swipe-card logs already carry that signal weeks earlier, in behaviour that precedes the formal finding?

Aims

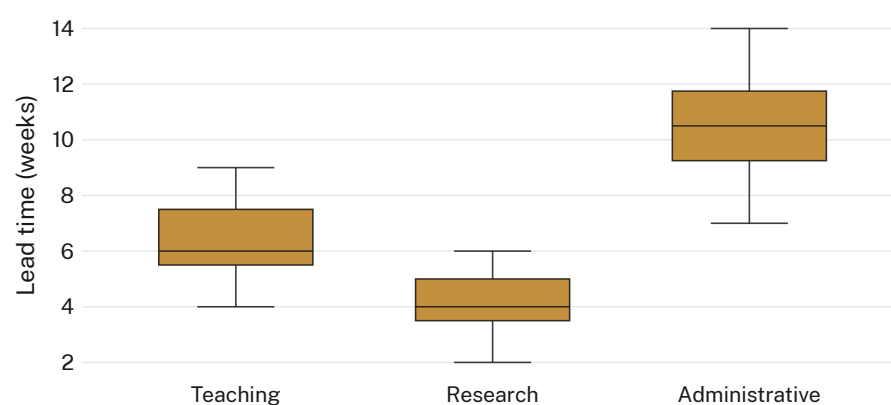
- Test whether booking-cancellation rates lead facilities' own utilisation reporting
- Build a swipe-card-derived quiet-building index
- Establish the earliest reliable lead time before a formal underuse finding

Methods

- 14 buildings · desk-booking + swipe-card logs · 5-min sampling · 18 months
- gradient-boosted hazard model on rolling occupancy features
- underuse threshold and lead-time window pre-registered

Results

The index flags underuse a median 6 weeks before facilities' quarterly review reaches the same building ($n = 14$, $r = 0.71$), with the lead widening for administrative floors.



Lead time between the index's alert and facilities' quarterly underuse finding, by building type ($n = 14$ buildings, 18-month trial, March–August 2025). Administrative floors lead by a median 10 weeks; research floors by 4.

Discussion & conclusions

Administrative floors lead furthest, consistent with a slower, more discretionary decline in use. Booking cancellations move first; badge data confirms it, and the two signals agree closely once a floor's quiet period exceeds three weeks. Next: extending the index to shared teaching precincts ahead of the 2026 winter re-audit, and testing whether a shorter sampling window shortens the lead time it reports.

References

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Desk-booking and swipe-card records de-identified at source; no individual movement traces retained; lead-time threshold pre-registered.

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